

Marina E. Nichols

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EDUCATION

Boston College, *Chestnut Hill, MA* Expected Dec 2025
Bachelor of Science in Applied Physics
Minor in Biology
GPA: 3.77

Tarleton State University, *Stephenville, TX* Jan 2022 - May 2023
Majored in Physics and Biomedical Sciences
GPA: 3.95
Distinctions: Honors College student, President's list, & Dick Smith Scholar

Howard Payne University, *Brownwood, TX* Aug 2021 - Dec 2021
GPA: 3.82
Distinctions: Honors College student

Cisco Junior College, *Cisco, TX* Aug 2019 - May 2021
GPA: 4.00

Bangs High School, *Bangs, TX* Aug 2017 - May 2021
GPA: 4.00
Distinctions: Salutatorian, Dual-Credit Student, NHS president

RESEARCH EXPERIENCE

Undergraduate Research Intern, *Massachusetts Institute of Technology* June 2025 - PRES
Microfluidics | *Device Optimization* | *CAD Software*
Mentor: Dr. Jongyoon Han

- Developed a recirculatory system for an inertia spiral microfluidic concentrator
- Optimized flow rate conditions for high throughput focusing of microalgae <10 μm
- Cultured algal and mammal cells in batches
- Met weekly as a group to discuss project progress and direction

Undergraduate Research Assistant, *Boston College* Sep 2023 - PRES
Biosensors | *Graphene Devices* | *Microfabrication* | *Data Analysis*
Mentor: Dr. Kenneth Burch

- Designed a microfluidic platform for sample delivery with SolidWorks
- Fabricated graphene field effect transistors (GFETs) for biosensing

- Performed viral protein detection in wastewater with multiplexed GFETs
- Met weekly in an individual or group setting to discuss project progress and direction

NSF Physics REU Researcher, *Texas Christian University*

May 2023 - Jul 2023

Biosensors | Quantum Dots | Optical Spectroscopy

Mentor: Dr. Anton Naumov

- Developed a COVID-19 biosensor with graphene quantum dots and ssDNA
- Characterized materials via spectroscopy (absorbance & fluorescence)

LEADERSHIP & PROFESSIONAL EXPERIENCE

Teaching Assistant, *Boston College*

Sep 2025 - PRES

- Coordinated with the professor and graduate TA to prepare applied physics labs
- Assisted undergraduate students during the lab

Residential Assistant, *Boston College*

Jan 2024 - May 2025

- Collaborated with the RA team to create & uphold community expectations
- Supported freshmen and sophomore students in the residence halls
- Conducted regular safety & maintenance checks

Animal Husbandry Assistant, *Animal Care Facility of Boston College*

Aug 2023 - Dec 2023

- Fed, cleaned, and cared for laboratory mice and rats in the morning
- Communicated important updates about animals to the staff and researchers

Supplemental Instructor, *Tarleton State University*

Aug 2022 - May 2023

- Created study materials for students in Physics I & II
- Led 1-hour study sessions twice a week to tutor undergraduate students

Veterinary Assistant, *Central Texas Veterinary Hospital*

May 2022 - Aug 2022

- Restrained and handled large and small animals in clinic and farm settings
- Prepared syringes, medications, charts, and samples for patients
- Learned sterile medical techniques and common diagnostic tests in the clinic

Team Leader, *Whataburger*

Jul 2020 - Jul 2022

- Prepared night shift schedule and position chart for 10-15 staff members
- Managed time, money, and inventory for the store during the evening shift
- Handled customer complaints and satisfaction, including reimbursements and discounts

Shift Leader, *Subway*

Jul 2019 - Jul 2020

- Opened and closed the store throughout the pandemic
- Managed money, inventory, and team members while on shift

- Communicated shift and inventory updates to upper management daily

PUBLICATIONS

Michael Geiwitz, Owen Page, Tio Mareello, **Marina E. Nichols**, et al. “Graphene Multiplexed Sensor for Point-of-Need Viral Wastewater-Based Epidemiology”, ACS Applied Bio Materials 7 (7), 4622-4632 (2024).

PRESENTATIONS

MIT Summer Research Program Research Symposium, Massachusetts Institute of Technology, August 2025. “Inertia Spiral Microfluidic Concentrator for Microalgae” (poster).

Society of Physics Students Poster Session, Boston College, November 2024. “Viral Protein Sensing in Wastewater with Graphene-Based Multiplexed Electronic Sensors” (poster).

Society of Physics Students Poster Session, Boston College, November 2023. “Graphene Quantum Dot-Based Biosensors for SARS-CoV-2 Detection” (poster).

REU Final Presentations, Texas Christian University, July 2023. “Graphene Quantum Dot-Based Biosensors for SARS-CoV-2 Detection”. (Oral)

TECHNICAL SKILLS

Fabrication: Photolithography, wire-bonding, e-beam deposition, soldering, 3D printing, CNC milling, and laser cutting

Computational: MATLAB, Python, SolidWorks, Fusion360